LAB 3: UML Class Diagram – Library Management System

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Aim

To identify the classes, attributes, methods and relationships from the requirements of the selected project (Library Management System) and to prepare the UML Class Diagram.

Theory

A class diagram is a structural UML diagram that shows the classes of a system, their attributes and methods, and the relationships between them. It is the main building block of object-oriented modelling.

Each class is drawn as a rectangle with three compartments:

- **Top compartment:** the name of the class.
- **Middle compartment:** the attributes (data) of the class.
- **Bottom compartment:** the methods (operations) of the class.

Visibility symbols used in the diagram: **+** means public and **-** means private.

The relationships used are:

- **Association:** a simple link between two classes, for example a Member borrows a Book.
- **Generalization (inheritance):** a child class inherits the attributes and methods of a parent class (is-a relationship). It is drawn with a hollow triangle at the parent end.
- **Multiplicity:** shows how many objects of one class can be linked to an object of the other class, for example 1, 0..1 or 0..*.

Project Requirements

The Library Management System should satisfy the following requirements:

- A member (student or faculty) can register, log in, search for books, reserve a book and pay fines.
- A librarian can add, update and delete books, issue books to members and accept returned books.
- An administrator can manage user accounts and generate reports.
- The system records every issue and return of a book along with the issue date, due date and return date.
- A fine is generated when a book is returned after the due date.
- A member can reserve a book that is currently issued to someone else.

Identification of Classes

The classes are identified by picking the nouns from the requirements. The verbs become the methods of the classes.

Class	Description
User	Parent class that holds the common data and login operations of all users of the system.
Member	A student or faculty member who borrows books from the library.
Librarian	Staff member who manages books and handles issue and return.
Administrator	Person who controls user accounts and reports of the system.
Book	A book of the library catalogue with its details and available copies.
BookIssue	A record of a book issued to a member, with issue date, due date and return date.
Reservation	A request made by a member to hold a book that is currently not available.
Fine	The penalty generated when a book is returned late.

Attributes and Methods

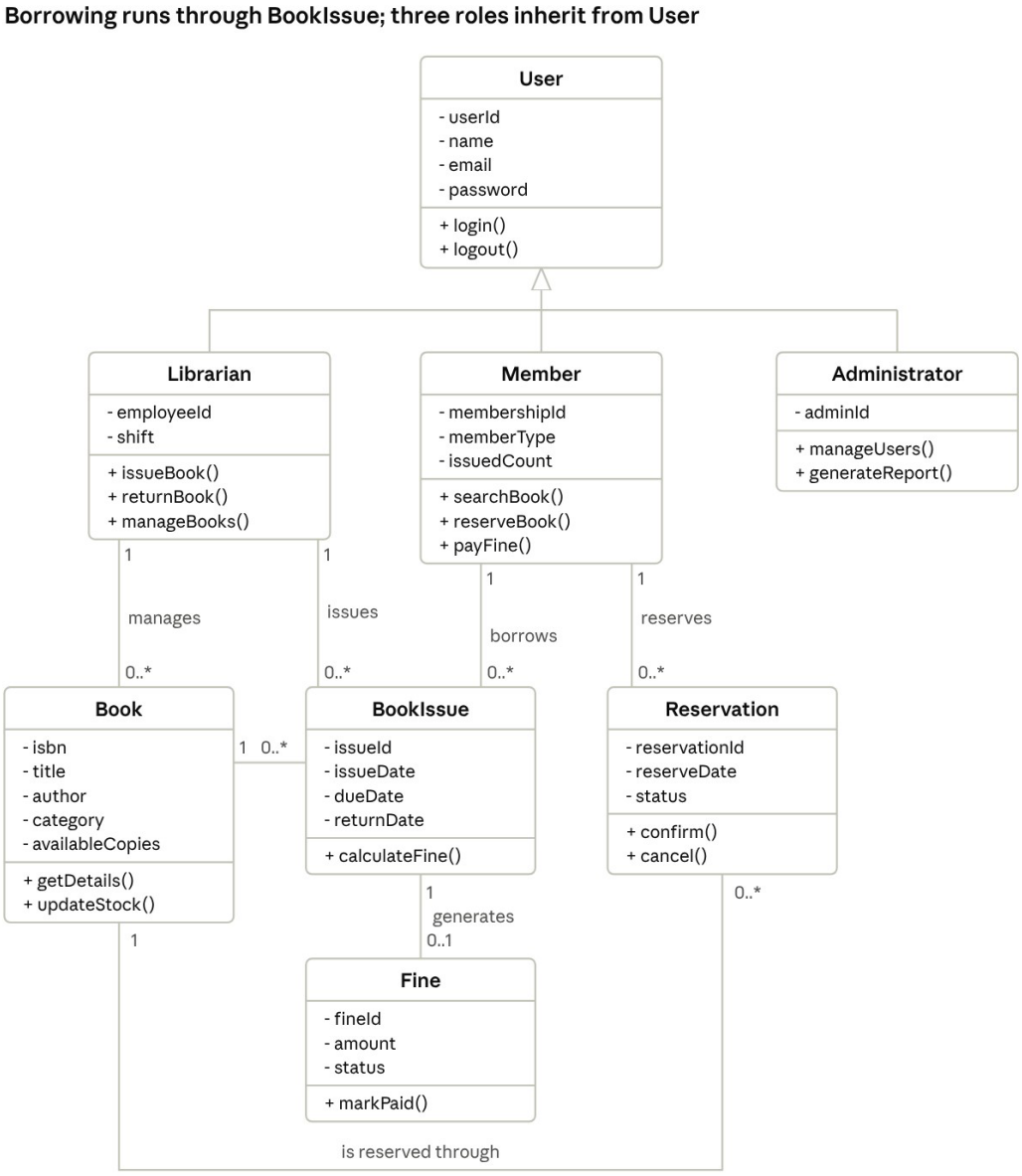
Class	Attributes	Methods
User	userId, name, email, password	login(), logout()
Member	membershipId, memberType, issuedCount	searchBook(), reserveBook(), payFine()
Librarian	employeeId, shift	issueBook(), returnBook(), manageBooks()
Administrator	adminId	manageUsers(), generateReport()
Book	isbn, title, author, category, availableCopies	getDetails(), updateStock()
BookIssue	issueId, issueDate, dueDate, returnDate	calculateFine()
Reservation	reservationId, reserveDate, status	confirm(), cancel()
Fine	fineId, amount, status	markPaid()

Relationships between Classes

Class A	Relationship	Class B	Multiplicity	Meaning
Class A	Relationship	Class B	Multiplicity	Meaning
Member, Librarian, Administrator	Generalization	User	–	Each of them is a User and inherits userId, name, email, password, login() and logout().
Member	Association	BookIssue	1 to 0..*	A member borrows many books over time.
Librarian	Association	BookIssue	1 to 0..*	A librarian issues and receives many books.
Librarian	Association	Book	1 to 0..*	A librarian manages (adds, updates, deletes) many books.
Member	Association	Reservation	1 to 0..*	A member can make many reservations.
Book	Association	BookIssue	1 to 0..*	A book can be issued many times.
Book	Association	Reservation	1 to 0..*	A book can be reserved many times.
BookIssue	Association	Fine	1 to 0..1	A late return generates at most one fine.

Class Diagram

The class diagram of the Library Management System is shown below. Member, Librarian and Administrator inherit from User, and the borrowing process is recorded through the BookIssue class.



class diagram · 8 classes, inheritance and associations

Figure 1: UML class diagram of the Library Management System. + means public, - means private.

Conclusion

The classes, attributes, methods and relationships of the Library Management System were identified from the project requirements and the UML class diagram was prepared. The diagram gives a clear structure of the system and works as a base for the coding and database design.